CS Notes (Groups)

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Outline

Introduction

Python

We will use Python in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the Spyder IDE.

Python fundamentals

The fundamental objects we need from Python are:

- lists: `[obj1, obj2, ...]`
- functions: `def func(x):  ...`

Creating Groups in Python

A simple group in Python

Suppose we want to model the coins group of the previous class. Then we will need to specify the *objects* and the *actions* of the group. We can use lists to hold the objects. . .

```
# group from 1.1 has as objects arrangements of a penny and a dime
arrangement1 = ['penny', 'nickel']
arrangement2 = ['nickel', 'penny']
objects = [arrangement1, arrangement2]
```

A simple group in Python

and functions to represent the actions:

```
# our action takes arrangement1 and turns it into arrangement2,
↪  and
# vice-versa
def group_action(arrangement):
    if arrangement[0] == 'penny':
        return arrangement2
    else:
        return arrangement1

# check that this does what we want it to
for object in objects:
    print(group_action(object))
```

```
['nickel', 'penny']
['penny', 'nickel']
```

Find the result of performing the group action twice in a row using the code above.
Answer: We can find the answer by computing `group_action(group_action(arrangement))` for the various arrangements...

Words group again

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!